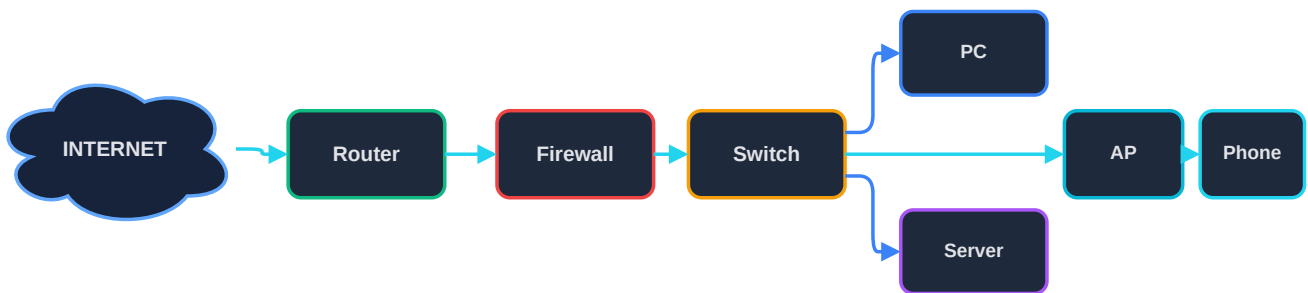


Networking Basics

Understanding How Devices Communicate



ANALOGY

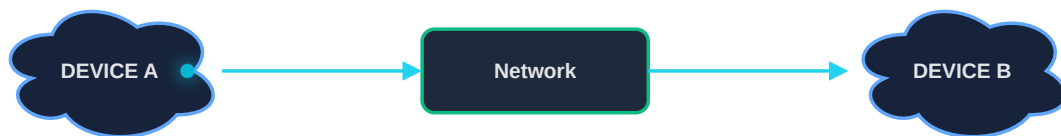
Think of this as learning the postal system before tracing one letter.

SPEAKER NOTES

Welcome to Networking Basics. Today we will explore how devices communicate, starting with the fundamental concepts and building up to a complete picture of how the internet works.

What Is Networking?

Device to Device via a Network



Data leaves as bits and returns as bits

ANALOGY

A network is like a road system: connections are roads, devices are destinations, protocols are traffic rules.

SPEAKER NOTES

Explain that networking underpins everything digital. A printer and laptop sharing a home router form a network even without the internet. Protocols are the agreed rules that make unlike devices work together. Ask: what does your phone do when it finds a printer?

Why Do We Need Networks?

What Networks Enable



Sharing



Internet



Files



Apps



Remote Work

One connection, many everyday jobs

ANALOGY

A shared network printer is like a community photocopier: one useful resource serves many people.

SPEAKER NOTES

Ask the room which networked service they used today. Point out that networks reduce duplicated equipment, enable immediate communication, and let applications run on remote servers.

Types of Networks

Scale of Networks



Each step connects more devices across larger distances

ANALOGY

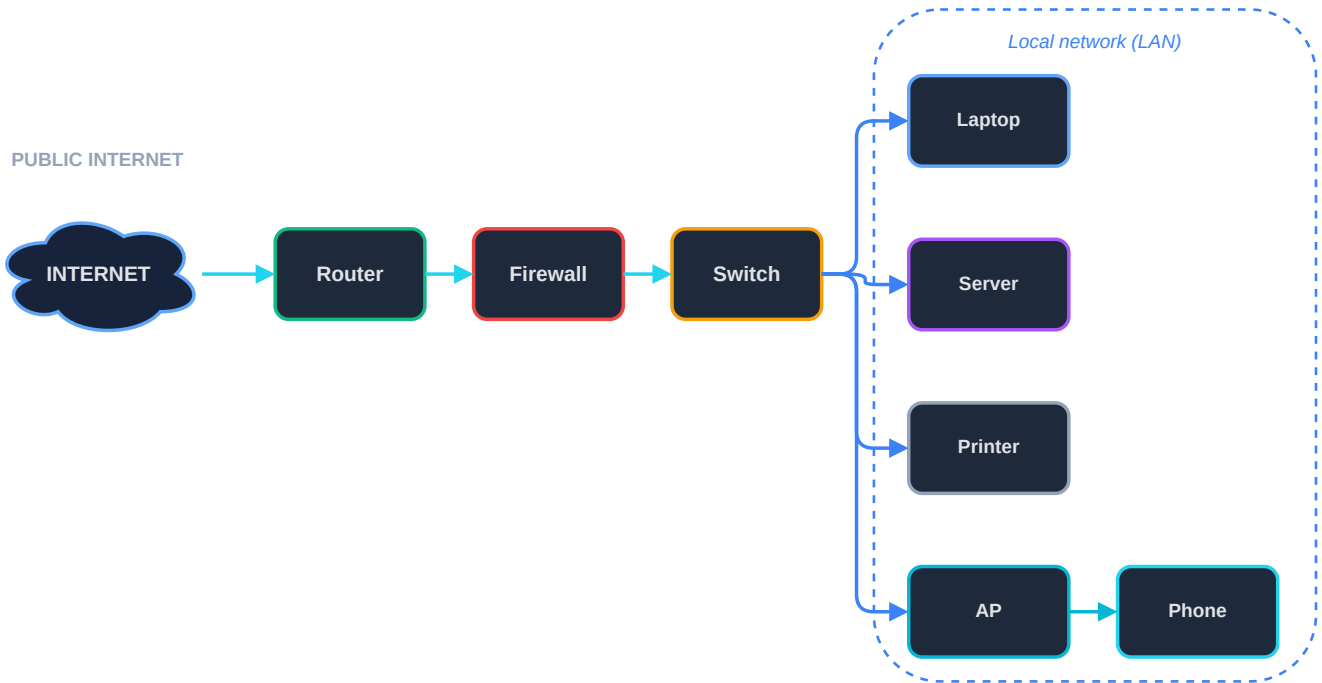
Each network type is like a circle of communication: your circle of friends (PAN), your neighborhood (LAN), your city (MAN), the whole country (WAN).

SPEAKER NOTES

Explain each scale with a relatable example. PAN is your watch syncing to your phone. LAN is your home wifi. WLAN is the same but wireless. MAN covers a city. WAN crosses countries. The internet is the ultimate WAN.

Network Components

From the internet to your devices



ANALOGY

A delivery needs a sender, a route, a carrier, rules, and a destination.

SPEAKER NOTES

Introduce the end-to-end path. Every box at home may combine multiple functions, but the jobs are distinct. Emphasize endpoints: servers are endpoints too; they provide services to clients.

Network Devices

Network Devices



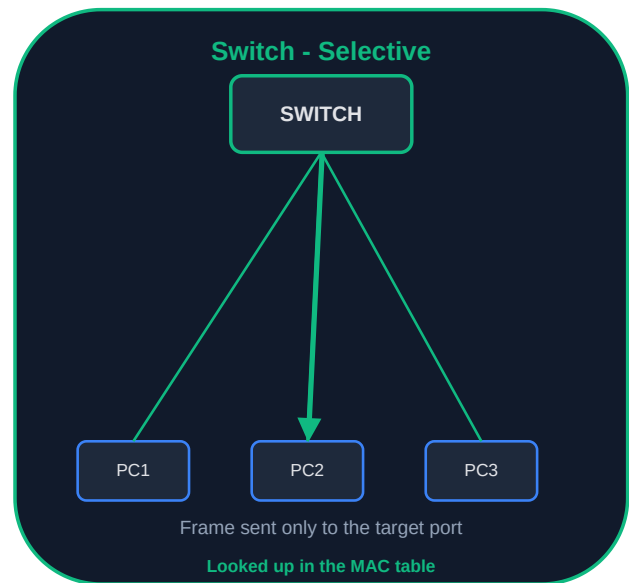
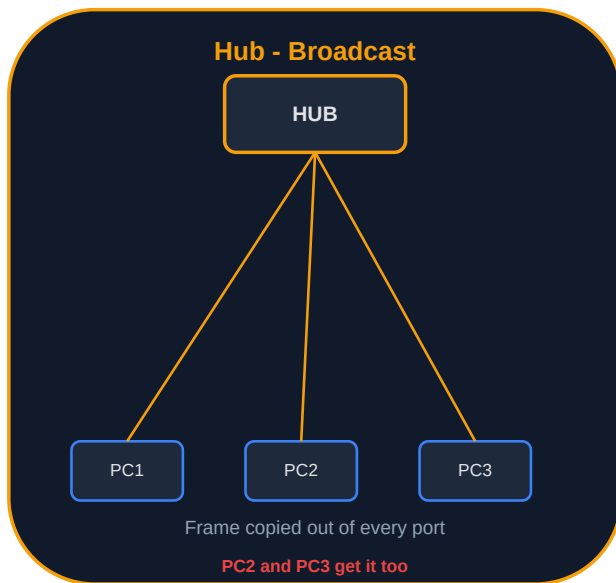
Same job, different layer of the stack

SPEAKER NOTES

Explain each device's role. A hub is like a megaphone - everything it hears, it shouts to everyone. A switch is like a smart receptionist - it knows where each device is and delivers only to the right recipient. A router is like a border guard - it knows the way to other networks.

Hub vs Switch

Hub vs Switch



This is why modern networks use switches, not hubs

ANALOGY

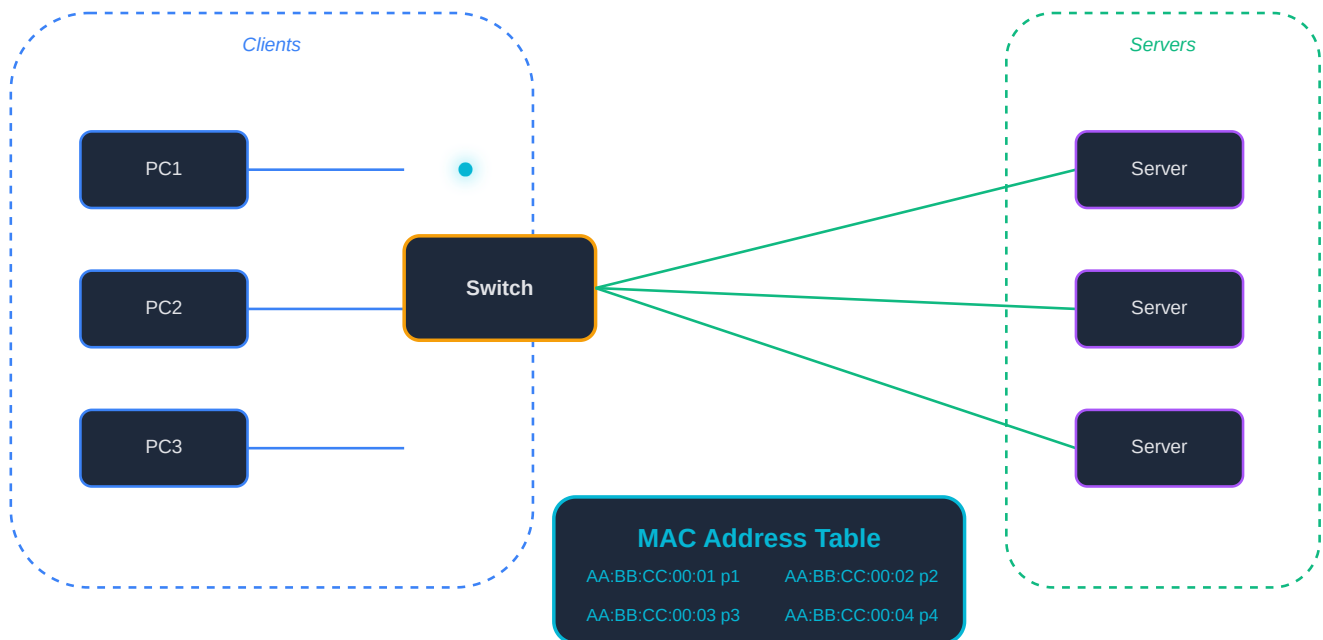
A hub is a town crier announcing to everyone. A switch is a receptionist delivering messages to specific offices.

SPEAKER NOTES

Demonstrate the broadcast vs selective forwarding. When PC1 sends to PC2, a hub sends to ALL ports. A switch learns MAC addresses and sends only to PC2. This is why modern networks use switches, not hubs.

Switching

How a Switch Learns



The switch learns which port each MAC lives behind

ANALOGY

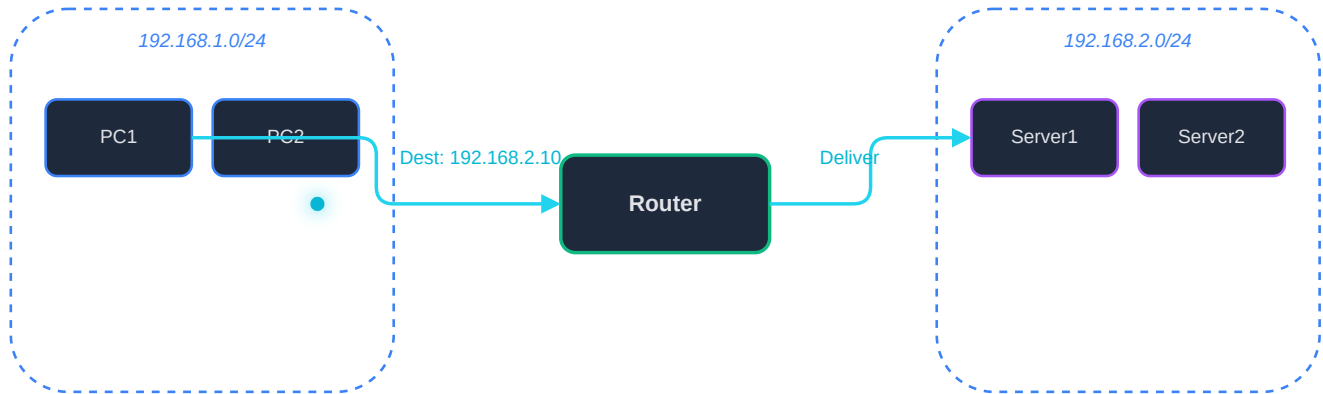
A switch learns like a doorman who remembers which badge belongs to which employee.

SPEAKER NOTES

Explain MAC address learning. When a switch receives a frame, it records the source MAC and the port. Then when it needs to send to a destination MAC, it looks up its table. If unknown, it floods (like a hub) until it learns. Explain the CAM table concept.

Routing

Inter-Network Routing



The router reads the destination IP, then forwards on the next hop - it never delivers the packet itself

ANALOGY

A router is like a city border guard: it checks the address on every packet and points it toward the right direction.

SPEAKER NOTES

Explain that routers strip the Layer 2 frame and make decisions based on IP headers. Each router in the path chooses the next hop. The default route catches unknown destinations. Routers on the same LAN exchange ARP to find each other's MAC addresses.

Router vs Switch

Switch vs Router



A router can do everything a switch does, and more

ANALOGY

A switch is a building concierge. A router is a postal sorting office.

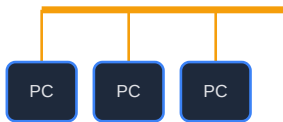
SPEAKER NOTES

Emphasize the fundamental difference: switches connect devices within a network, routers connect networks themselves. In home routers, both functions are combined. Enterprise networks separate these roles for better control.

Network Topologies

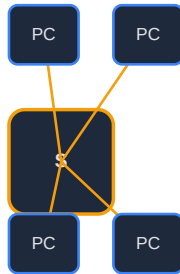
Network Topologies

Bus



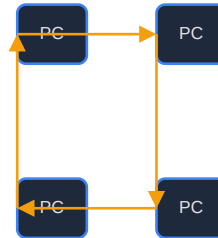
One shared cable

Star



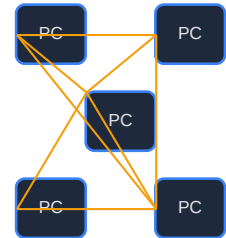
Central hub or switch

Ring



Each node has two links

Mesh



Redundant, no single point of failure

ANALOGY

Network topologies are like city planning: grid streets, ring roads, star-shaped suburbs.

SPEAKER NOTES

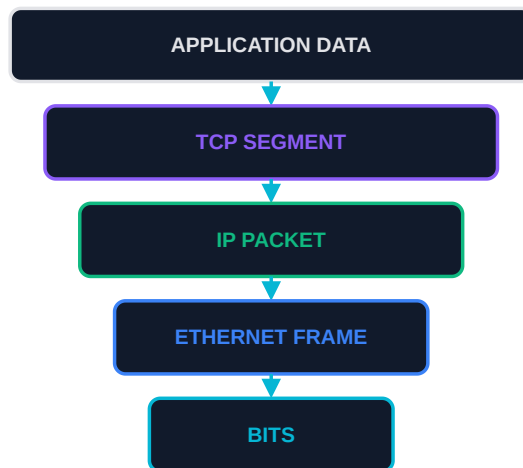
Each topology has tradeoffs. Bus is simple but a break kills the segment. Star is reliable but the switch is a single point of failure. Ring has predictable latency. Mesh offers redundancy. Tree scales well. Modern networks are usually star-of-stars (hierarchical) with redundant links.

Data Encapsulation

Data Encapsulation (Sending)

Sender

Receiver



Headers are added at each layer going down, stripped going up

ANALOGY

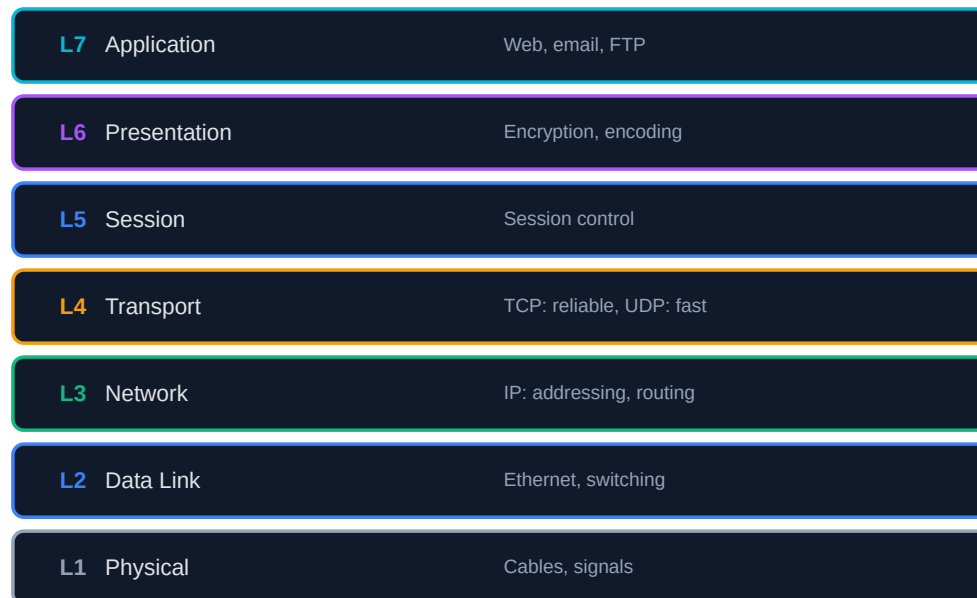
A letter is written, protected in an envelope, routed by city and street, carried over successive roads, then unwrapped at each level.

SPEAKER NOTES

Walk down from the browser and then up at the server. The local Wi-Fi frame is addressed to the gateway's MAC when the server is remote. Routers unwrap the incoming link frame and create another frame for the next link. HTTPS/TLS crosses the simple OSI boundaries.

OSI Model

OSI Model - 7 Layers



ANALOGY

The OSI layers are like a shipping manifest: each layer adds or checks information as the package moves through the system.

SPEAKER NOTES

The OSI model is a teaching tool. It maps functions to seven layers. In practice, TCP/IP does not follow it exactly, but the concepts are still useful. Layers 1-2 are physical media, layer 3 is logical addressing, layers 4-7 handle end-to-end communication.

OSI Model In Action

OSI Model in Action

Sender



Receiver

Data goes down the sender stack and up the receiver stack

ANALOGY

These are the carrier, city address, local delivery label, and the physical road or signal.

SPEAKER NOTES

Devices can operate at multiple layers. A modern Layer 3 switch routes as well as switches. A firewall may inspect IPs, ports, connection state, and application data. The simple layer associations are a first map, not a rule that every product fits one box.

TCP/IP Model

TCP/IP Model (4 Layers)



The practical four-layer model the internet actually runs

ANALOGY

TCP/IP is the practical toolkit; OSI is the textbook map. Both describe the same territory in different ways.

SPEAKER NOTES

The TCP/IP model is what the internet actually runs. Unlike OSI's seven layers, it has four. Application combines OSI layers 5-7. Transport and Internet map directly. Link covers OSI layers 1-2. This is the model developers and engineers work with daily.

MAC vs IP Addresses

MAC vs IP Addresses

MAC Address

A4:34:0F:12:AB:91

Physical, burned into hardware

Never changes

Only valid on the local segment

Forwarded by switches

IP Address

IPv4: 192.168.1.25

IPv6: 2001:db8::25

Logical, assigned by DHCP or manually

Changes with the network

Routable across the internet

Forwarded by routers

ARP bridges a MAC address to an IP address on the local network

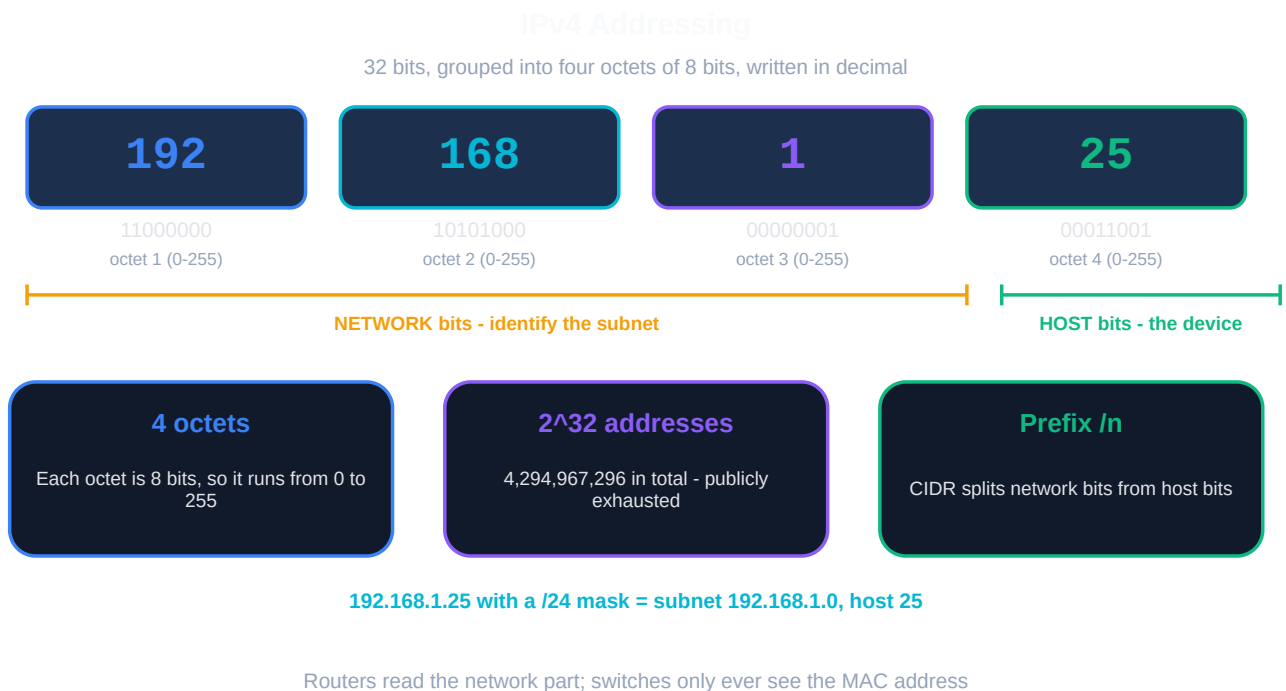
ANALOGY

A MAC address is like a device serial number. An IP address is like a temporary desk assignment that can change.

SPEAKER NOTES

MAC addresses are burned into hardware or randomly generated for privacy. They are only meaningful on the local network segment. IP addresses are assigned by DHCP or manually and are meaningful across the entire routing path. ARP bridges the two by discovering MAC addresses for local IPs.

IPv4 Addressing



ANALOGY

An IP address is like a seat number on a train: the carriage part tells you the train, the seat part tells you exactly where to sit.

SPEAKER NOTES

Walk the diagram left to right: 192.168.1.25 is four octets, and under each one is its 8-bit binary form. Point at the two brackets - the amber one covers the network bits and the green one the host bits, and the /24 mask is what decides where the split happens. Contrast with the MAC slide: the IP address is logical and can change, the MAC is fixed to the hardware and only works locally.

IP Address Classes

IP Address Classes

The leading bits of the first octet once decided how big the network was

Class	Starts	First octet	Default mask	Usable hosts	Typical use
A	0	1 - 126	255.0.0.0 (/8)	16,777,214	Very large networks
B	10	128 - 191	255.255.0.0 (/16)	65,534	Medium business networks
C	110	192 - 223	255.255.255.0 (/24)	254	Small networks, most homes
D	1110	224 - 239	no subnet mask	n/a	Multicast group addresses
E	1111	240 - 255	no subnet mask	n/a	Experimental, reserved

Classes are obsolete - CIDR (/n) sizes networks today, but these ranges still explain why 10.x and 192.168.x are private

ANALOGY

Classes are like vehicle sizes: a bus seats many, a car seats a few, and the number plate's first digit told you which one you were looking at before the label was written on the window.

SPEAKER NOTES

Teach the rule first, then the history. Read the first octet in binary: a leading 0 means Class A, 10 means Class B, 110 means Class C, 1110 is Class D (multicast) and 1111 is Class E (reserved). 192 starts with 110, so every 192.x address is Class C. Stress that classes were retired in the 1990s because they wasted addresses - that is exactly why CIDR exists - but the numbering is why 10.0.0.0/8 and 192.168.0.0/16 are the private blocks.

IP Address Ranges

IPv4 Address Ranges

Whole blocks of IPv4 are set aside - never assign these on a public network

This network 0.0.0.0/8 Refers to the local network itself	Private (Class B) 172.16.0.0/12 Covers 172.16 to 172.31
Private (Class A) 10.0.0.0/8 Largest private block, 16.7M addresses	Private (Class C) 192.168.0.0/16 Covers 192.168.0 to 192.168.255
Loopback 127.0.0.0/8 ping localhost always answers here	Multicast 224.0.0.0/4 One-to-many delivery, e.g. IPTV
APIPA 169.254.0.0/16 Self-assigned when DHCP fails	Broadcast 255.255.255.255 Every device on this subnet

Private ranges are reused in every home and office - the router translates them to one public IP

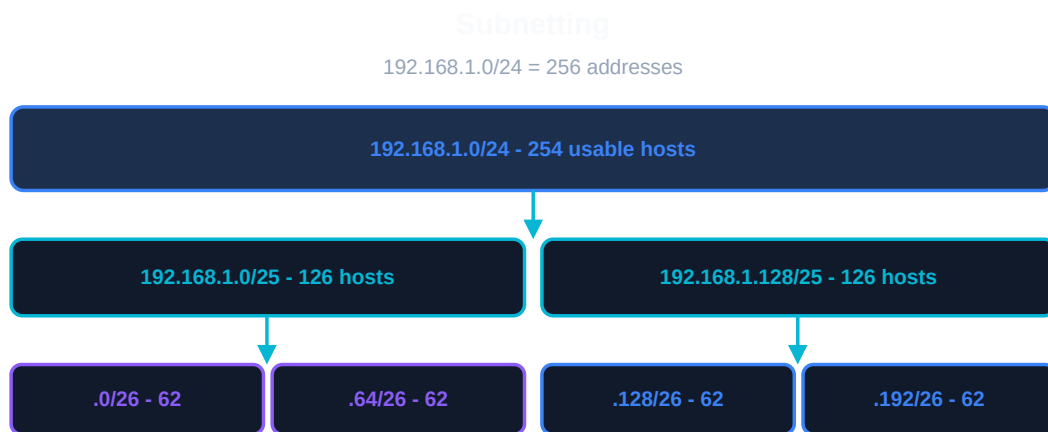
ANALOGY

Private addresses are extension numbers inside a building: everyone can have extension 101, but only the building's main street address is unique in the city.

SPEAKER NOTES

These eight blocks are the ones worth memorising. Loopback explains why ping localhost always works even with no cable plugged in. APIPA is the 169.254.x.x address students see when a DHCP server is unreachable - it is the classic symptom of a DHCP failure. The three private blocks answer the question 'why can my laptop be 192.168.1.5 and so can my neighbour's?' because those addresses are never routed on the public internet; NAT translates them at the router.

Subnetting



More subnets means better isolation and tighter security

ANALOGY

Subnetting is like dividing a big parking lot into sections so traffic flows better and you know which section each car belongs to.

SPEAKER NOTES

Start with 192.168.1.0/24 and split into 192.168.1.0/25 and 192.168.1.128/25. Each subnet has its own broadcast domain. Subnetting improves security and reduces broadcast traffic.

TCP vs UDP

TCP vs UDP

TCP

- Connection-oriented
- Reliable delivery
- Ordered data
- Acknowledgement
- Flow control
- Congestion control

HTTPS, SSH, FTP

UDP

- Connectionless
- Best effort
- No acknowledgement
- Low overhead
- Fast delivery
- No flow control

DNS, Streaming, Gaming

Choose TCP when nothing may be lost, UDP when speed wins

ANALOGY

TCP is a phone call where you wait for the answer. UDP is shouting across a crowd. Fast, but not guaranteed.

SPEAKER NOTES

TCP is like certified mail with a signature. It ensures every packet arrives in order. UDP is like a postcard - fast but no guarantee. Use TCP for web pages and file transfers. Use UDP for streaming.

Common Protocols

Common Protocols			
	Protocol	Port	Purpose
●	HTTP	:80	Web (unencrypted)
●	HTTPS	:443	Web (encrypted)
●	DNS	:53	Names to IPs
●	DHCP	:67/68	Auto IP assignment
●	TCP	:-	Reliable transport
●	UDP	:-	Fast transport
●	IP	:-	Logical addressing
●	ICMP	:-	Diagnostics (ping)

ANALOGY

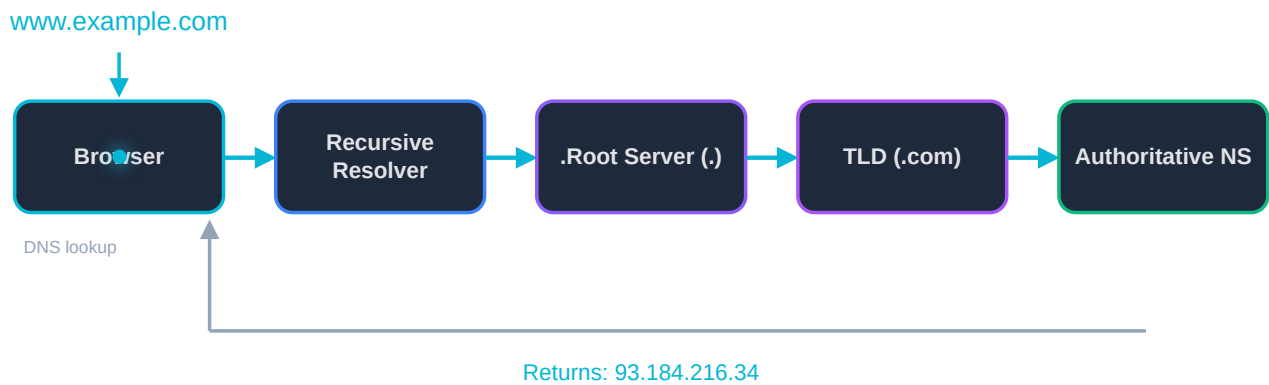
Protocols are like languages: each layer speaks its own language, using the one below it as a carrier.

SPEAKER NOTES

Show the protocol stack as a layered diagram. HTTP and HTTPS are application layer. TCP and UDP are transport. IP is network layer.

DNS

DNS Resolution



ANALOGY

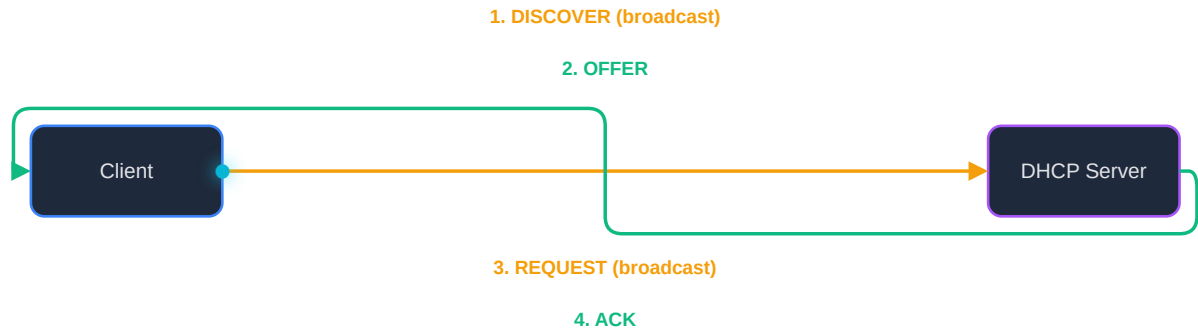
DNS is like calling directory assistance: you ask for a name, they look up the phone number.

SPEAKER NOTES

Animate the resolution: User enters **www.example.com** -> local DNS resolver -> root server -> **.com** TLD server -> authoritative NS -> final IP. DNS typically uses UDP port 53, but TCP for large responses.

DHCP

DHCP - DORA Process



The client broadcasts, the server answers - then the lease is locked in

ANALOGY

DHCP is like a hotel front desk: a guest arrives, asks for a room, gets an offer, confirms, and receives a key with an expiration date.

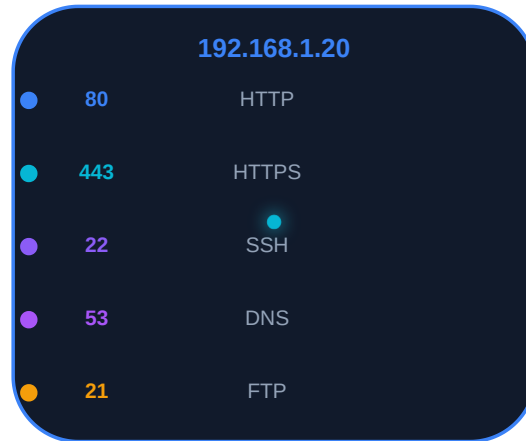
SPEAKER NOTES

Animate the four-step DORA process with message arrows between client and server. Mention that DHCP also provides DNS server addresses, default gateway, and subnet mask. If no offer arrives the client falls back to APIPA, which links back to the IP ranges slide.

Ports & Sockets

Ports & Sockets

An IP address is a building; ports are doors



A socket is a source IP:port plus a destination IP:port

192.168.1.20:443

ANALOGY

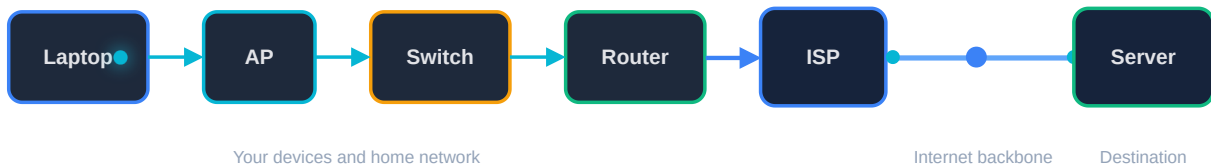
An IP is a building address. A port is a door number. A socket pair is the full mailing label.

SPEAKER NOTES

Think of an IP address as a building address. Ports are individual apartment doors. A server can have HTTP on port 80, HTTPS on 443, SSH on 22. The combination of source and destination IP+port uniquely identifies a connection.

How the Internet Works

How the Internet Works



Each hop forwards your data one step closer to the server

ANALOGY

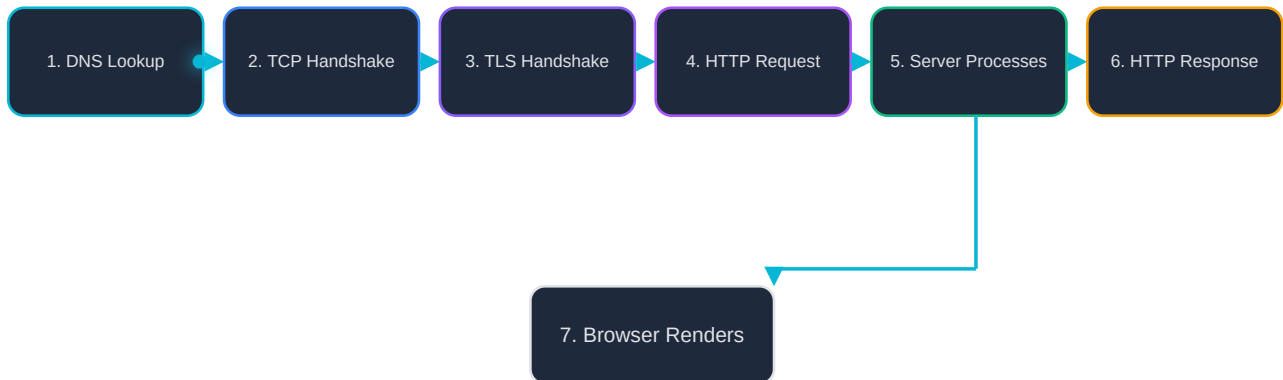
The internet is like a global postal system: each post office only knows the next stop, but the system delivers anywhere.

SPEAKER NOTES

Trace: Laptop -> Wi-Fi -> AP -> Switch -> Router -> ISP -> Internet Backbone -> Destination ISP -> Destination Server. Each hop is a routing decision based on the destination IP.

Opening a Website

Loading a Website



Six round trips before the first pixel is painted

ANALOGY

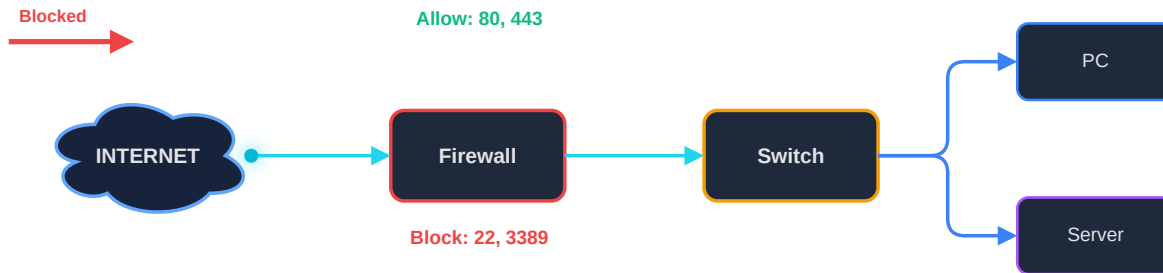
Opening a website is like ordering at a restaurant: look up the menu (DNS), signal readiness (TCP), verify authenticity (TLS), place order (GET), receive food (response), enjoy (render).

SPEAKER NOTES

Start with typing `https://example.com`. DNS resolves to IP. TCP SYN/SYN-ACK/ACK handshake. TLS negotiates cipher suites. HTTP GET request. Server responds. Browser renders.

Firewall & Security

Firewall & Network Security



Every packet is inspected against a rule set before it passes

ANALOGY

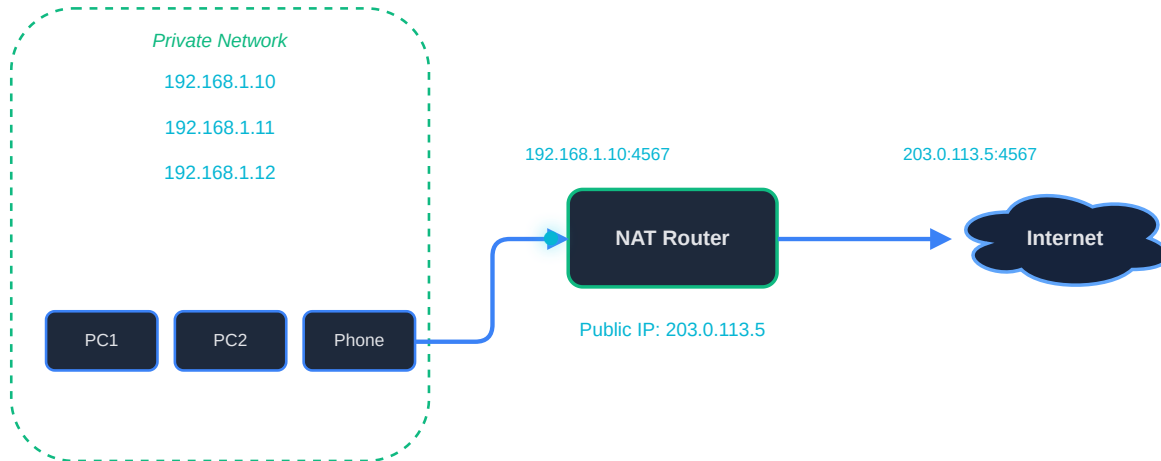
A firewall is like a security guard: checking IDs, admitting invited guests, turning away suspicious visitors.

SPEAKER NOTES

Firewalls inspect packets against rules. They can allow port 443 but block port 22. Packet filtering at L3, stateful inspection at L4, deep packet inspection at L7. Next-gen firewalls add application awareness.

NAT

NAT - Network Address Translation



One public IP, many private devices - all reachable from outside

ANALOGY

NAT is like a building with one street address but many apartments. The doorman knows which apartment each letter belongs to.

SPEAKER NOTES

NAT extends the life of IPv4. Home router has one public IP, many private. Router replaces source IP with public IP outbound, and translates back on return. Inbound connections require port forwarding. This is the slide that closes the loop on the private ranges we saw earlier.

Network Security

Network Security Layers



Defence in depth: no single layer has to be perfect

ANALOGY

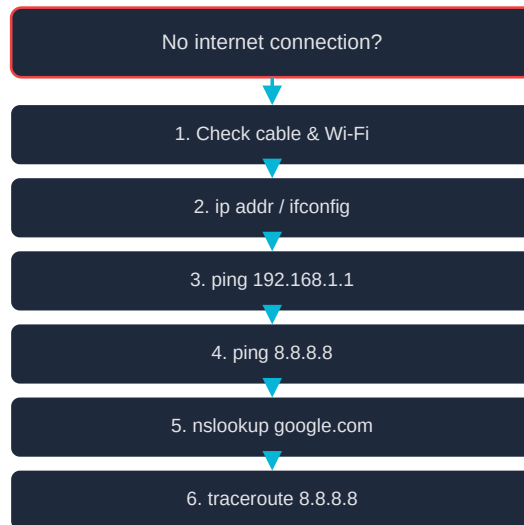
Network security is like securing a house: fence (firewall), safe (encryption), separate rooms (segmentation), alarm system (IDS/IPS).

SPEAKER NOTES

Security is layered - defense in depth. Firewall is perimeter. VPN extends private network. IDS monitors; IPS blocks. Segmentation limits blast radius. Encryption protects data in transit.

Troubleshooting

Troubleshooting Flow



Work outward: link, local IP, gateway, internet, DNS

ANALOGY

Troubleshooting is like diagnosing a car: check power (cable), fuel (IP), engine (gateway), road (internet path).

SPEAKER NOTES

If no internet: 1) Is cable plugged in? Check Wi-Fi. 2) ip addr - has IP? 3) ping 192.168.1.1 - if fails, local issue. 4) ping 8.8.8.8 - if fails, ISP issue. 5) nslookup google.com - if fails, DNS issue.

Networking Cheat Sheet

Networking Cheat Sheet



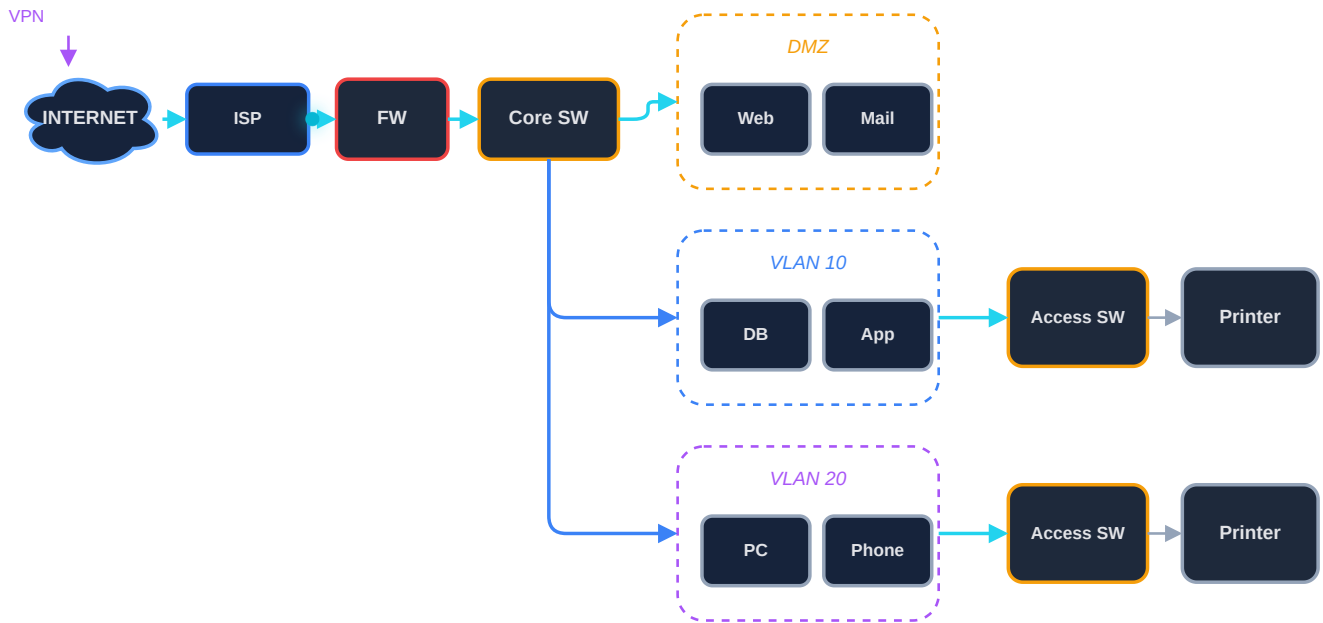
If you remember one thing per slide, remember these

SPEAKER NOTES

Review: MAC is local. IP routes across networks. Switch = Layer 2. Router = Layer 3. TCP reliable, UDP fast. DNS names to IPs. DHCP assigns IPs. Firewalls protect boundaries.

Final Network Map

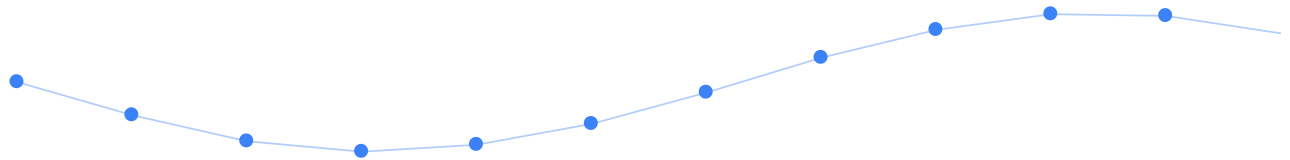
Complete Network Architecture



SPEAKER NOTES

Complete enterprise architecture: Internet -> ISP (BGP) -> Firewall -> DMZ (servers) -> Core Switch -> VLANs -> Access Switches -> End devices. VPN connects remote users. This ties together everything we learned.

Questions



Questions

Thank you

ANALOGY

The best network engineers never stop asking questions. Each answer leads to deeper understanding.

SPEAKER NOTES

Open floor for questions. Encourage students to ask about anything confusing. These fundamentals form the foundation for all digital communication.

Networking Basics — Study Guide

Read this before you present. It explains **why** each slide says what it says, so you can explain the material rather than read it off the screen. Slide numbers match `js/slides.js` (34 slides, ids 0–33).

The one-sentence story of the whole deck:

> Two devices talk over a shared medium; the **MAC address** gets the frame to the > right device on *this* link, the **IP address** gets the packet to the right > *network*, the **layers** make both possible, and **DNS, DHCP, TCP and NAT** > turn that machinery into something a person can use.

Part 1 — The 20-second version

If a question comes and you need to answer it fast:

Question	Answer
Hub vs switch?	Hub copies every frame to every port. Switch learns MAC→port and forwards only to the target.
Switch vs router?	Switch = Layer 2, MAC, inside one LAN. Router = Layer 3, IP, between networks.
MAC vs IP?	MAC is burned into hardware, changes never, only valid locally. IP is assigned, routable everywhere.
OSI vs TCP/IP?	OSI is 7 layers for teaching. TCP/IP is 4 layers and is what actually runs.
TCP vs UDP?	TCP = reliable, ordered, acknowledged (web, files). UDP = fast, no guarantee (video, gaming, DNS).
What is a port?	A number identifying <i>which door</i> on a machine. IP finds the building, port finds the door.
DNS / DHCP?	DNS = name → IP. DHCP = automatically hands out IPs (DORA).
Why NAT?	IPv4 ran out. NAT lets many private devices share one public IP.
What is a subnet mask?	It marks which bits are network and which are host: <code>/24</code> = 24 network bits + 8 host bits.

Part 2 — Slide by slide

1. Title — Networking Basics (slide 1)

Point at the diagram: Internet → Router → Firewall → Switch → PC/Server → AP → Phone. Say:
"This is the whole journey in one picture. Every device we meet today sits on this path."

2. What Is Networking? (slide 2)

- A network is **two or more devices agreeing to exchange data under shared rules**.
- The rules are **protocols**. Without protocols, a printer and a laptop cannot understand each other.
- Key nuance: **a network does not need the internet**. A laptop + printer + home router is already a network.
- Today's journey: **YOUR DEVICE → LOCAL NETWORK → INTERNET → SERVICE**

3. Why Networks? (slide 3)

Four reasons, in the order people care about: **share resources → communicate → use remote services → manage centrally**. Ask the audience which networked service they used today. It always lands better than a definition.

4. Types of Networks (slide 4)

Scale from smallest to largest — this ordering is the whole point of the slide:

Type	Stands for	Example
PAN	Personal Area Network	Phone ↔ smartwatch, phone ↔ earbuds
LAN	Local Area Network	Home or office wifi, one building
WLAN	Wireless LAN	The same LAN but over radio
MAN	Metropolitan Area Network	A city-wide network, a university campus
WAN	Wide Area Network	Across countries/continents; the internet is the largest WAN

5. Network Components (slide 5)

End-to-end: **Internet → Router → Firewall → Switch → endpoints (PC, server, AP, phone)**. Home routers combine several of these boxes into one device; the *jobs* are still separate.

6. Network Devices (slide 6)

Device	Layer	Job	Analogy
Hub	1	Repeats to every port	Town crier shouting
Switch	2	Forwards frames to the target port	Receptionist
Router	3	Joins different networks	Border guard
Access Point	2	Bridges wired ↔ wireless	Translator
Modem	1	Digital ↔ analog signal	Phone line adapter
Firewall	3-7	Filters traffic by rules	Security guard

7. Hub vs Switch (slide 7) — the single most important comparison

- **Hub:** PC1 sends to PC2 → the hub copies the frame out **every** port. PC2 *and* PC3 receive it. Wasteful and a security leak.
- **Switch:** PC1 sends to PC2 → the switch looks up the MAC table, finds PC2's port, and delivers **only** there.
- **Why it matters:** hubs waste bandwidth and leak traffic; switches do not. Every modern LAN is switches.

8. Switching (slide 8) — how the switch learns

The mechanism is a *source-address learning loop*:

1. A frame arrives on port 3 from MAC `AA:BB:CC:00:01`.
2. The switch writes `AA:BB:CC:00:01 → port 3` into its **CAM/MAC table**.
3. For the destination it looks up the table.
4. **If the MAC is unknown**, it floods the frame (like a hub) *until* someone answers — **unknown unicast flooding**.

Say the punchline: *"The switch learns from the source, and forwards using the destination."*

9. Routing (slide 9)

- A router **strips the Layer 2 frame**, reads the **destination IP**, and consults its **routing table** to pick the next hop.
- Every router repeats that decision. The packet walks hop by hop toward the destination.
- If no route matches, the **default route (0.0.0.0/0)** sends it onward.
- On the same LAN, routers find each other's MAC addresses using **ARP**.

10. Router vs Switch (slide 10)

The clean rule: **a switch connects devices; a router connects networks**. A switch that sees a destination on another network forwards the frame to its gateway — it does not route.

11. Topologies (slide 11)

Topology	Advantage	Disadvantage
Bus	Cheapest, least cable	One break kills the whole segment
Star	Easy to add/remove a device	The central switch is a single point of failure
Ring	Predictable timing, can be dual-ring	A break affects the whole ring unless redundant
Mesh	Very resilient, no single point of failure	Expensive — lots of cabling
Tree/Hierarchical	Scales well	Depends on the core it grows from

Real networks are **star-of-stars**: a hierarchical tree of stars with redundant uplinks.

12. Encapsulation (slide 12) — why layering works

Sending:

Application Data → TCP Segment (adds ports) → IP Packet (adds IPs)
→ Ethernet Frame (adds MACs) → Bits on the medium

Receiving reverses it — each layer strips *its own* header and passes the rest up. **This is what makes the internet possible:** a Wi-Fi device and a fibre device can share the same IP packet because they only have to understand IP.

13. OSI Model (slide 13)

#	Layer	Unit	Example
7	Application	Data	HTTP, DNS, SSH
6	Presentation	Data	TLS, encoding, compression
5	Session	Data	Session setup/teardown
4	Transport	Segment	TCP, UDP — ports
3	Network	Packet	IP, ICMP, ARP — logical addressing
2	Data Link	Frame	Ethernet, Wi-Fi — MAC addressing
1	Physical	Bits	Cable, radio, fibre

Mnemonic: "**All People Seem To Need Data Processing.**"

14. OSI in Action (slide 14)

Trace one HTTPS request down the sender's stack and up the receiver's:

Browser data → TLS + TCP 443 → IP packet to the server → Wi-Fi frame to the gateway → radio bits

Note honestly: real products straddle layers (a Layer 3 switch routes *and* switches; a firewall may inspect IP, port, state and application data).

15. TCP/IP Model (slide 15) — the real one

OSI 7,6,5 → TCP/IP Application (HTTP, DNS, SSH, TLS)
OSI 4 → TCP/IP Transport (TCP, UDP)
OSI 3 → TCP/IP Internet (IP, ICMP, ARP)
OSI 2,1 → TCP/IP Link (Ethernet, Wi-Fi, physical)

Four layers instead of seven. **Say this plainly: OSI is the teaching map, TCP/IP is the road.**

16. MAC vs IP (slide 16) — memorize this contrast

	MAC	IP
Nature	Physical (hardware)	Logical (assigned)
Example	A4:34:0F:12:AB:91	192.168.1.25 / 2001:db8::25
Changes?	Never (unless spoofed)	Changes with the network
Scope	Local segment only	Routable across the internet
Forwarded by	Switches	Routers

ARP is the bridge: "which MAC sits behind this IP?" — answered on the local segment only.

17. IPv4 Addressing (slide 17)

- IPv4 = **32 bits**, written as **four octets** in dotted-decimal: 192.168.1.25 .
- Each octet = 8 bits → **0 to 255**. Under each octet on the diagram is its binary form.
- Total space: $2^{32} = 4,294,967,296$ addresses.
- The **two brackets** on the diagram are the key visual: amber = **network bits**, green = **host bits**. The **subnet mask (/n) decides where the split happens**.
- Worked example: 192.168.1.25/24 → network 192.168.1.0 , host 25 .
- Rule of thumb: a /24 gives 256 total, **254 usable** (first = network, last = broadcast).

18. IP Address Classes (slide 18)

Teach the rule with binary. Read the first octet:

Class	Leading bits	First octet	Default mask	Usable hosts	Use
A	0	1-126	255.0.0.0 (/8)	16,777,214	Very large networks
B	10	128-191	255.255.0.0 (/16)	65,534	Medium business
C	110	192-223	255.255.255.0 (/24)	254	Small networks, most homes
D	1110	224-239	—	—	Multicast
E	1111	240-255	—	—	Experimental/reserved

Worked check: **192 = 11000000**, which starts 110 → **Class C**. Every 192.168.x.x address is Class C.

The history matters: classes were retired in the 1990s because a Class C always gave you 254 addresses whether you needed 3 or 250 — hugely wasteful. **CIDR (/n) replaced them** so networks can be any size. But the class numbering is exactly *why* 10.0.0.0/8 and 192.168.0.0/16 are the private blocks.

19. IP Address Ranges (slide 19)

The eight blocks worth memorising:

Range	Name	Meaning
0.0.0.0/8	This network	Refers to the local network
10.0.0.0/8	Private (Class A)	Largest private block, ~16.7M addresses
127.0.0.0/8	Loopback	Every device answers for itself — why <code>ping localhost</code> always works, even unplugged
169.254.0.0/16	APIPA	Self-assigned because DHCP failed
172.16.0.0/12	Private (Class B)	172.16 → 172.31
192.168.0.0/16	Private (Class C)	192.168.0 → 192.168.255
224.0.0.0/4	Multicast	One-to-many delivery (IPTV)
255.255.255.255	Broadcast	Every device on this subnet

The three private ranges are the ones that matter most. They are reused in *every* home and office on earth — they are never routed on the public internet. That answers "*why can my laptop be 192.168.1.5 and so can my neighbour's?*" NAT (slide 29) translates them to one public IP.

APIPA is the practical one: see a `169.254.x.x` address on a PC → the DHCP server could not be reached. Link it forward to the DHCP slide.

20. Subnetting (slide 20)

- A subnet splits one network into smaller ones. The **mask decides the split**: `/24` = 24 network bits + 8 host bits.
- Worked example: `192.168.1.0/24` → two `/25` s: `192.168.1.0/25` and `192.168.1.128/25`. Each becomes its own **broadcast domain**.
- Why do it: **smaller broadcast domains** (less wasted traffic) and **better isolation** (security).

21. TCP vs UDP (slide 21)

	TCP	UDP
Connection	Oriented (handshake)	Connectionless
Delivery	Reliable, ordered, acknowledged	Best-effort
Speed	Slower overhead	Fast
Control	Flow + congestion control	None
Use	Web pages, email, file transfer	Video calls, gaming, DNS

Analogy: **TCP is certified mail with a signature; UDP is a postcard.**

22. Common Protocols (slide 22)

Protocol	Port	Transport	Purpose
HTTP	80	TCP	Web browsing
HTTPS	443	TCP	Encrypted web
DNS	53	UDP (TCP for large)	Name → IP
DHCP	67/68	UDP	Automatic IP assignment
SSH	22	TCP	Encrypted remote shell
FTP	20/21	TCP	File transfer
SMTP	25	TCP	Sending email
IP	—	—	Addressing + routing
ICMP	—	—	ping, traceroute

23. DNS (slide 23)

Name → IP, done as a chain of referrals:

```
you → local resolver → root server → .com TLD server → authoritative nameserver → IP
```

Results are **cached**, which is why the second visit is instant.

24. DHCP (slide 24) — DORA

1. Discover	"Is there a DHCP server out there?"	(broadcast)
2. Offer	"I can give you 192.168.1.42"	(broadcast)
3. Request	"I'll take 192.168.1.42"	(broadcast)
4. Acknowledge	"It's yours"	(unicast)

DHCP also delivers the **subnet mask, default gateway and DNS servers** — not just the IP. Leases expire and must be renewed. If no server answers → **APIPA** (slide 19).

25. Ports & Sockets (slide 25)

- **IP = building address. Port = door number. Socket pair = full mailing label.**
- One server can offer many services by listening on many ports (80, 443, 22...).
- A connection is uniquely identified by **source IP + source port + destination IP + destination port**.

26. How the Internet Works (slide 26)

```
Laptop → Wi-Fi → AP → Switch → Router → ISP → Backbone → Destination ISP → Server
```

- The **default gateway** is the router that everything not-local is sent to.
- Each router independently reads the destination IP and picks a next hop.

- **Autonomous Systems (ASes)** are run by the same organisation; **BGP** is the protocol that exchanges routes between them.

27. Opening a Website (slide 27) — put it all together

1. DNS lookup	example.com → 93.184.x.x
2. TCP handshake	SYN → SYN-ACK → ACK
3. TLS handshake	agree encryption keys, verify the certificate
4. HTTP request	GET / HTTP/1.1
5. Response	200 OK + HTML
6. Render	browser draws HTML, then loads CSS/JS/images

Every earlier slide appears here. This is the payoff slide for the whole deck.

28. Firewall (slide 28)

Rules decide allow/block — e.g. **allow 443**, **block 22**. Escalating depth: packet filtering (L3) → stateful inspection (L4) → deep packet inspection (L7).

29. NAT (slide 29) — the payback for slide 19

```
Outbound: 192.168.1.25 : 51000 → 203.0.113.9 : 51000 (source rewritten, port kept)
Return:   203.0.113.9 : 51000 → 192.168.1.25 : 51000 (the NAT table remembers the mapping)
```

- Many private devices share **one** public IP. That is how IPv4 survives.
- **Inbound** connections have no existing mapping, so they need **port forwarding**.

30. Network Security (slide 30)

Defence in depth — no single control is enough: **Firewall** (perimeter) · **VPN** (encrypted tunnel over untrusted networks) · **IDS/IPS** (detect / block) · **Segmentation** (VLANs limit the blast radius) · **Encryption** (protects data in transit).

31. Troubleshooting (slide 31) — work outward

1. Physical	Is the cable in? Is Wi-Fi on?
2. IP config	Do I have an address? (ip addr / ifconfig)
3. Gateway	ping 192.168.1.1 → fails = local problem
4. Internet	ping 8.8.8.8 → fails = ISP / routing problem
5. DNS	nslookup google.com → fails = DNS problem
6. Path	tracert → shows where it stops

Each step eliminates a layer, so you always know *where* the problem is.

32. Cheat Sheet (slide 32)

Rapid recall of the whole deck — use this slide as your revision sheet.

33. Final Network Map (slide 33)

Ties everything together: Internet → ISP (BGP) → Firewall → DMZ servers → Core switch → VLANs → Access switches → End devices, with **VPN** for remote users.

34. Questions (slide 34)

Part 3 — How to explain it (delivery notes)

The narrative arc. Don't present 34 disconnected facts. The deck has a spine: > *Devices need to talk (2-6) → so we build addressing (7-10, 16-19) → the > addressing works because we layer it (11-15) → the protocols make it useful > (20-27) → then we protect it (28-30) → and finally we debug it (31).*

The three comparisons are the heart of it. Hub vs switch, switch vs router, TCP vs UDP. If the audience remembers only three slides, make it these.

Every analogy in the deck, in one list:

Slide	Analogy
What is networking	Roads, destinations, traffic rules
Why networks	Community photocopier
Network types	Circles of friends → neighbourhood → city → country
Network components	Sender, route, carrier, rules, destination
Hub vs switch	Town crier vs receptionist
Switching	Doorman remembering badges
Routing	Border guard checking addresses
Router vs switch	Building concierge vs postal sorting office
Topologies	City planning
Encapsulation	Letter in envelope, via city and street
OSI	Shipping manifest
TCP/IP	Practical toolkit vs textbook map
MAC vs IP	Serial number vs desk assignment
IPv4 addressing	Train seat number
IP classes	Bus vs car, read from the number plate
IP ranges	Extension numbers inside a building
Subnetting	Sections of a parking lot
TCP vs UDP	Certified mail vs a postcard
Protocols	Layered languages
DNS	Directory assistance
DHCP	Hotel front desk
Ports/sockets	Building, door number, mailing label
Internet path	Global postal system
Opening a website	Ordering at a restaurant
Firewall	Security guard
NAT	One street address, many apartments
Security	Fence, safe, separate rooms, alarm
Troubleshooting	Diagnosing a car

Anticipated questions, with short answers.

Question	Short answer
Why is the MAC 6 groups of 2 hex digits?	6 × 8 bits = 48 bits, the Ethernet hardware address size.
What exactly is an octet?	One group of 8 bits. "Octet" = eight. In IPv4 there are four.
Is 127.0.0.1 the same as localhost?	Yes — localhost is resolved to 127.0.0.1 via the hosts file.
What is the difference between /24 and 255.255.255.0?	Nothing. 255.255.255.0 is the dotted mask, /24 is its shorthand.
Why is the first and last host address unusable?	First = network ID, last = broadcast address.
Can two devices have the same IP?	Not on the same subnet — that is an IP conflict and breaks both.
What is ARP?	"Which MAC is behind this IP?" Asked locally, answered by whoever owns it.
Is HTTPS TCP or UDP?	TCP (HTTP/3 is the exception — it runs on QUIC/UDP).
Why does a switch need a MAC table?	Without it a switch would flood like a hub and waste every port.
Does a router use MAC addresses?	Only to reach the next hop on its own local segment. It routes on IP.

Pacing. Roughly 45–60 minutes for 34 slides. The heavy slides are 7 (hub vs switch), 8 (switching), 17–19 (the addressing block), 21 (TCP vs UDP) and 27 (opening a website). If you are short on time, expand those and skim the rest — they are the ones people remember.

If someone asks about IPv6, the short version: IPv6 uses **128 bits**, written in hex, and the space is so large that **private ranges are no longer needed** — every device can have a globally unique address, which is exactly what NAT was working around.

Part 4 — Working on the deck

```
cd networking-presentation

node scripts/generate-drawio.js # rebuild networking-diagrams.drawio (34 pages)
node scripts/export-svgs.js    # export pages -> assets/diagrams/*.svg + diagram-svgs.js
```

Every diagram is authored in **draw.io** — there are no hand-drawn SVGs left. Open [networking-diagrams.drawio](#) in draw.io to edit any diagram; the page order matches the slide order, and each page is a fixed 800×460 canvas.

Where	What
<code>js/slides.js</code>	Slide titles, bullets and speaker notes (ids must stay 0-33)
<code>scripts/generate-drawio.js</code>	Builds the <code>.drawio</code> source; <code>PAGES</code> at the bottom sets page order
<code>scripts/export-svgs.js</code>	<code>PAGES</code> maps page number → diagram name → SVG file
<code>css/animations.css</code>	Diagram motion (connector dash flow, breathing glow)

Controls: ← → or **Space** to move, **F** for fullscreen, **Home/End** to jump, the notes button (bottom right) shows the speaker notes for the current slide.

> If you add a slide, add its `id` in sequence, add a diagram page in > `generate-drawio.js`, and add the matching row in **both** `PAGES` tables — > `export-svgs.js` fails loudly if a page is missing.